



EchoAgent AI

Transform Voice into Actionable Workflow



01

Overview

EchoAgent AI captures real-time audio input, transcribes it via AssemblyAI / Web Speech Engine, processes the intent using structured prompts, and executes the right task — turning your voice into actionable results.



Voice Input
(Real-time Audio)



Transcription
(AssemblyAI / Web Speech)



Intent Processing
(LLM + Prompts)



Task Execution
(Actionable Workflow)

02

System Architecture



Audio Input
(Real-time)



Transcription
(AssemblyAI /
Web Speech Engine)



Intent Processing
(Structured Prompt
+ LLM)



Task Execution
(Actions / Workflow)

01 Project Overview

What EchoAgent AI does and why it matters

From Voice → Intent → Action

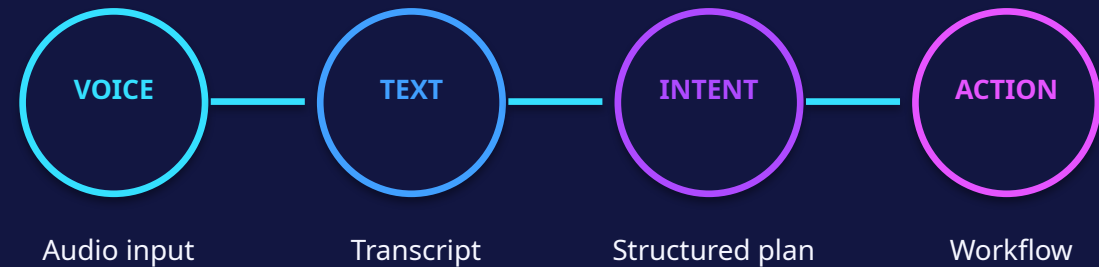
EchoAgent AI turns spoken instructions into structured, actionable workflows—without requiring users to manually type, organize, or dispatch every task.

Real-time speech capture & transcription

Context-aware intent parsing

Automated task / reminder / alert generation

Live UI state and checklist synchronization



Outcome: a voice request becomes a usable workflow state.

02 System Architecture

Layered architecture for capture, reasoning, and action dispatch

1

Audio Capture Layer

Web Speech API / audio streamer
Real-time waveform visualization

2

Transcription Engine

Speech → text stream
Context / mode aware input

3

Intent Parsing & Action Planner

Structured prompt patterns
Decisions • tasks • priorities • payloads

4

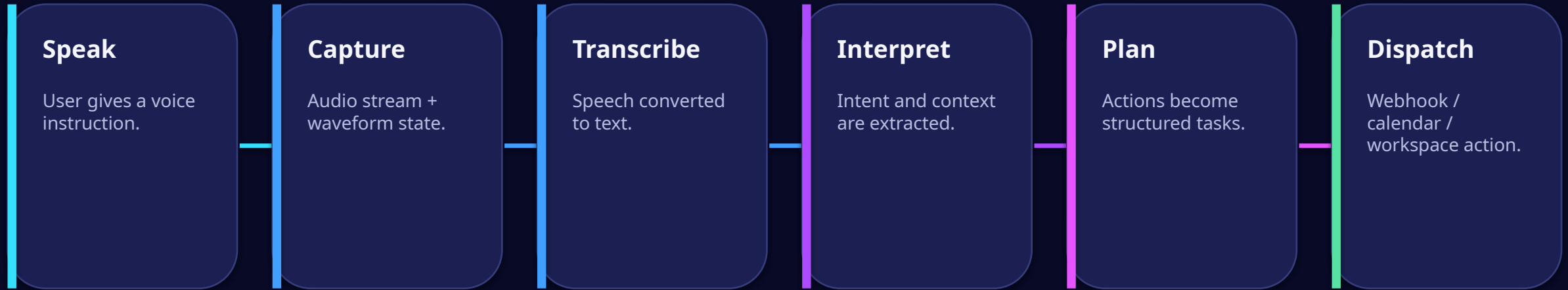
UI Response State

Live status indicators
Dynamic checklist updates

Modular • event-driven • extensible

03 End-to-End Workflow

How a spoken request moves through EchoAgent AI



Key principle: every stage exposes a clear state so the UI can stay synchronized in real time.

04 Context-Aware Modes

The same voice interface adapts to different operational contexts

Meeting Mode

VOICE → STRUCTURE

- Extract key decisions
- Map action items
- Assign priority tags
- Generate calendar reminder payloads

SRE Incident Mode

VOICE → STRUCTURE

- Detect incident severity
- Create quick remediation tasks
- Prepare webhook alerts
- Keep response state visible

Content Strategy Mode

VOICE → STRUCTURE

- Generate social/newsletter drafts
- Structure publishing tasks
- Prepare Notion workspace updates
- Track completion state

05 Component Structure

Core frontend files and their responsibilities

App.jsx

Main State Orchestration

Audio event listeners
Speech / transcript state
Mode selection and action planning
Dynamic checklist and status updates

index.css

Visual System & Motion Layer

Tailored glassmorphism theme
Dark blue / purple visual language
Custom keyframe waveform animations
Responsive UI presentation

06 Technology & Integrations

Key technologies supporting the EchoAgent AI pipeline



Web Speech API

Real-time speech capture



AssemblyAI

Transcription engine option



Structured Prompts

Intent + action planning



Webhooks

External alert / action
dispatch



Calendar Payloads

Reminder automation



Notion Updates

Workspace synchronization

07 Example Voice-to-Action Scenarios

What a user can accomplish with a single spoken request

MEETING

"Create tasks for the three decisions we just made and remind me tomorrow."

OUTPUT • Decisions → prioritized tasks → calendar payload

SRE

"This incident is critical. Prepare remediation tasks and alert the on-call webhook."

OUTPUT • Severity → remediation checklist → webhook

CONTENT

"Turn this idea into a social post and add it to my content workspace."

OUTPUT • Draft → publishing task → workspace update

The interface keeps the user focused on intent—not manual data entry.

08 Future Scope

Natural extensions for a production-ready agent

Multi-agent orchestration

Separate specialist agents for meetings, SRE, and content workflows.

Persistent memory

Remember project context, recurring preferences, and previous actions.

More integrations

Slack, Jira, Trello, Google Calendar, Notion, and custom APIs.

Voice feedback

Spoken confirmation, interruption handling, and richer conversational state.

09 Project Summary

A concise value proposition for judges, reviewers, and contributors

EchoAgent AI

Transform Voice into Actionable Workflow

Captures and transcribes voice in real time

Understands context and converts intent into structured actions

Automates tasks, reminders, alerts, and workspace updates

Provides a clear, responsive state-driven user experience

Built for hackathons • extensible for real-world automation